Twintec TVTI Portal

Full Technical & Functional Specification Report

Comprehensive overview covering Architecture, Role Portals, Practical Slot Management, Security Hardening, Dynamic Watermarking, and Recent UI/UX Enhancements.

Project: TVTI Mobile & Web App **Date:** November 8, 2026 **Branch:** `ifthi` (Clean & Verified)

Backend: Node.js / Express / MongoDB **Mobile:** React Native / Expo **Web:** React / Vite / Tailwind

1. Executive Summary & Core Objectives

The Twintec Vocational Training Institute (TVTI) platform is a unified cross-platform management ecosystem. It provides dedicated role-based portals for **Students**, **Instructors**, and **Administrators**. Key recent milestones include complete hardening against screen capture and recording, unification of modular video lectures and PDF study materials, intelligent practical slot scheduling with multi-tier conflict resolution, automated real-time notifications, and glassmorphic responsive authentication screens.

2. System Architecture & Tech Stack

Tier	Technologies	Key Responsibilities
Mobile Client	React Native 0.74+, Expo 51+, React Navigation, Reanimated, BlurView	Native iOS & Android mobile application with biometric & responsive web support.
Web Frontend	React 18, Vite, Tailwind CSS, Lucide Icons, Axios	High-performance web management and student portal for desktop browsers.
Backend API	Node.js, Express, TypeScript, Mongoose, Socket.io, Multer	RESTful API with role-based auth middleware, rate limiting, and real-time socket events.
Database & Storage	MongoDB Atlas, Cloudinary CDN, Local Protected Storage	Encrypted user records, session scheduling, media streaming, and PDF materials.
Security Engine	Android FLAG_SECURE, Expo ScreenCapture, DOM Blur Shield	Zero-leakage video streaming, watermark overlays, and DevTools/recording interception.

3. Role-Based Portals & Functional Modules

A. Student Portal

Production Ready

- **Dashboard & Home:** Live announcements, upcoming scheduled practical sessions, attendance metrics, and enrolled course overview.
- **Unified Video Lectures & PDF Notes:** Modular grouping (Module 1, 2, etc.) where video players and downloadable/viewable PDF lecture notes sit together inside the module accordion.
- **Practical Slot Booking:** Browse open slots by date, view real-time available seats, 1-tap booking, and cancellation request workflow.
- **Academic Results:** Semester exam grades, practical evaluation scorecards, and instructor feedback logs.

B. Instructor Portal

Production Ready

- **Practice Sessions Management:** Create slots with custom location, equipment notes, capacity, and instructor assignment. Enforces a strict 1-slot-per-day instructor policy.
- **Upload Management:** Publish YouTube URLs or direct video files, attach PDF notes, and upload standalone course documents.
- **Student Evaluation:** Grade practical attendance, mark completion status, and review cancellation requests.
- **Announcements:** Broadcast batch-specific or global news items directly to student mobile notifications.

C. Administrator Portal

Production Ready

- **Master Slot Scheduling:** Multi-view calendar (Day, Week, Month, and List view) with slot locking, capacity adjustments, and instructor reallocation.
- **Course & Batch Management:** Create batches, assign lead instructors, and manage course syllabi.
- **User Management:** Approve pending student registrations, assign index numbers, trigger password resets, and manage role privileges.

4. Advanced Screen Recording & Content Protection

To preserve proprietary vocational training curriculum, a multi-layer anti-piracy system was developed:

Hardware & Operating System Level

Security Hardened

- **Android (FLAG_SECURE):** Configured native window flags that force black screens when Android OS screen recorder, external capture cards, or third-party apps attempt to capture the viewport.
- **iOS (ExpoScreenCapture):** Native iOS recording detection that immediately hides the media layer and replaces video playback with a security blackout shield.
- **AppState Transition Guard:** Video player auto-pauses and dims when mobile applications are minimized or switched to split-screen/pip mode.

Desktop & Web Capture Defense

Security Hardened

- **Screen Capture API Blocking:** Trapped `navigator.mediaDevices.getDisplayMedia` and `getDisplayMediaWithStream` to reject unauthorized web capture requests.
- **Shortcut Interception:** Blocks `PrintScreen`, `F12`, `Ctrl+Shift+I` (Inspect Element), and `Ctrl+Shift+S` with automatic visual warnings.
- **Window Focus Polling:** 400ms interval polling checks `document.hasFocus()`. If OS snipping tools or Game Bar overlays take focus, the video element is blanked out immediately.
- **Moving Anti-Piracy Watermark:** Displays dynamic student identity data (Student Name, Registration/Index Number, Timestamp) floating randomly across the video feed every 4 seconds, rendering screen recordings forensically traceable.

5. Detailed Summary of Recent Upgrades & Fixes

Area / Component	Problem Addressed & Implementation Details	Files Affected
Student & Instructor Modules	Unified module cards so videos and PDFs sit inside the same card box (`moduleSectionBox`), matching Instructor layout.	`VideosScreen.tsx` `UploadVideoScreen.tsx`
Practical Slot Management	Enforced 1-slot-per-day limit per instructor; blocked booking/editing on expired or past dates; unified Day/Week/List views.	`AdminSlotManagementScreen.tsx` `PracticeSessionsScreen.tsx` `practiceSlotController.ts`
Auth Screen Theme	Removed redundant "Back to Sign In" links; centered card titles; converted Force Password Change to desktop 2-column glassmorphism.	`LoginScreen.tsx` `RegisterScreen.tsx` `ForgotPasswordScreen.tsx` `ForceChangePasswordScreen.tsx`
Slot Cancellation Workflow	Allowed instructors/admins to approve or reject student slot cancellations with instant real-time student notification updates.	`studentPracticeController.ts` `NotificationsScreen.tsx`
Date & Time Pickers	Resolved horizontal overflow on small devices by stacking Hour and Minutes vertically with clean scroll pickers.	`DateTimePickerModals.tsx`

6. Enhanced Entity-Relationship (EER) Architecture

The TVTI Portal relies on a structured relational data model on MongoDB. Below is the Entity-Relationship matrix mapping primary entities, relations, and cardinalities:

Parent Entity	Relationship	Child Entity	Cardinality	Description
USER	Specialization	**STUDENT / INSTRUCTOR / ADMIN**	`1 : 0..1`	Disjoint role specialization based on `role` enum.
COURSE	Offers	**BATCH**	`1 : M`	One course schedules multiple batches over time.
BATCH	Enrolls	**ENROLLMENT**	`1 : M`	A batch holds multiple student enrollments.
STUDENT	Enrolls in	**ENROLLMENT**	`1 : M`	Students can enroll in multiple training batches.
BATCH	Assigns	**INSTRUCTOR**	`M : N`	Multiple instructors assigned via `instructor_ids[]`.
BATCH	Hosts	**PRACTICE_SLOT**	`1 : M`	Batches schedule weekly practical lab sessions.
INSTRUCTOR	Supervises	**PRACTICE_SLOT**	`1 : M`	Strictly limited to 1 practical slot per day.
PRACTICE_SLOT	Reservations	**SLOT_BOOKING**	`1 : M`	Students reserve lab seats up to slot capacity.
STUDENT	Books	**SLOT_BOOKING**	`1 : M`	Unique compound index prevents duplicate bookings.
BATCH	Content	**VIDEO (Lecture/PDF)**	`1 : M`	Unified video lectures and PDF study documents.
STUDENT	Attendance	**ATTENDANCE**	`1 : M`	Logs presence, absence, late, or excused status.
STUDENT	Evaluations	**RESULT**	`1 : M`	Records marks and grades for course assessments.
COURSE	Applications	**APPLICATION**	`1 : M`	Prospective student admissions and payment review.

7. Quality Assurance, Build & Verification Status

- **TypeScript Compilation:** `npx tsc --noEmit` passes with **0 errors** across all backend and mobile TypeScript files.
- **API Stability:** Express 5 backend verified with route guards, input validation, and secure password hashing via `bcryptjs`.
- **Git Cleanliness:** Working directory is clean, synchronized with upstream branch `ifthi`, and compliant with commit constraints.